# A1.2 Nucleic acids

*Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-a1-2.html*

*Written for this site from the class slides. The chapter is on pages 14–24 of the course textbook. Image credits are on the last slide of the slides for this subtopic.*

## BUILDING BLOCKS

### A1.2.1 DNA is the genetic material of life

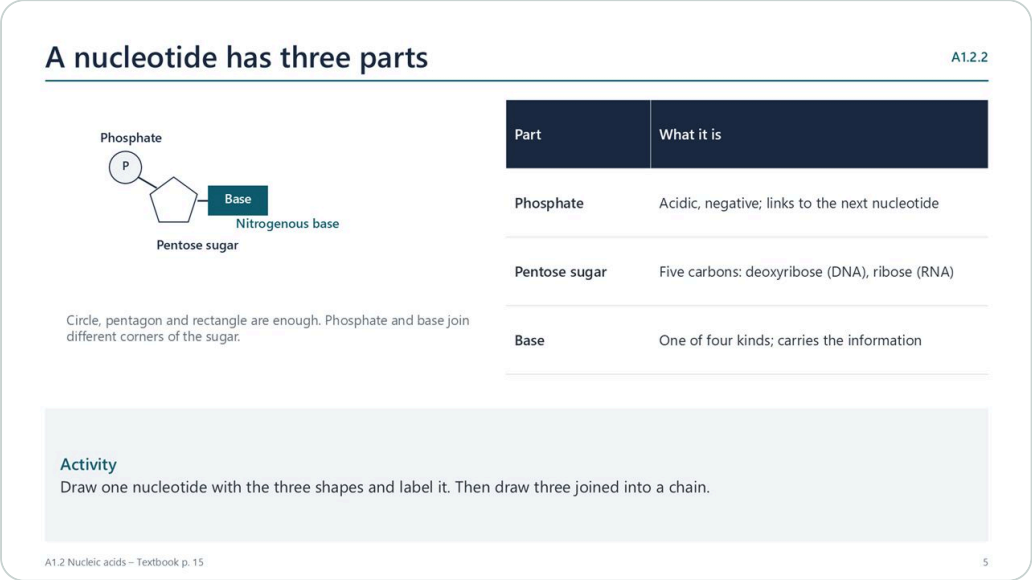
pp. 14–24

All living organisms use DNA as their genetic material. Some viruses use RNA, but viruses are not considered to be living.

### A1.2.2 The three parts of a nucleotide

pp. 14–24

A nucleotide has a **phosphate**, a **pentose sugar** and a **nitrogenous base**. In diagrams, use a circle, a pentagon and a rectangle, in these positions.




### A1.2.3 The sugar-phosphate backbone

pp. 14–24

Each sugar bonds to the phosphate of the next nucleotide. This makes a continuous chain of covalently bonded atoms, so that the backbone is strong. Any base can follow any other.

### A1.2.4 Bases carry the code

pp. 14–24

DNA has **adenine, cytosine, guanine** and **thymine**. RNA has **uracil** in place of thymine. Only the base varies along a strand, so the base sequence carries the information.

### A1.2.5 RNA is a polymer formed by condensation

pp. 14–24

RNA nucleotides are monomers that join by **condensation**, releasing water for each new bond. RNA is a single strand with a sugar-phosphate backbone.

## DNA

### A1.2.6 The DNA double helix

pp. 14–24

Two **antiparallel** strands, running in opposite directions, are linked by **hydrogen bonds** between complementary bases: A with T, and G with C.

> **Exam tip:** draw the two strands antiparallel. You do not need to draw the helix, or to remember the numbers of hydrogen bonds.

### A1.2.7 Differences between DNA and RNA

pp. 14–24

| | DNA | RNA |
|---|---|---|
| STRANDS | Two | One |
| SUGAR | Deoxyribose | Ribose |
| BASES | A, C, G, T | A, C, G, U |

Ribose has an -OH group on the carbon where deoxyribose has only -H, so deoxyribose has one oxygen atom fewer.

## USING THE STRUCTURE

### A1.2.8 Why base pairing matters

pp. 14–24

Pairing depends on hydrogen bonds, which form only between the correct bases. Each strand can therefore act as a template in **replication**. In **expression**, DNA is a template for mRNA, and anticodons on tRNA pair with codons on mRNA.

### A1.2.9 Limitless capacity to store information

pp. 14–24

Any length of DNA molecule and any base sequence is possible. With four bases, there are 4 possible sequences for one position, 16 for two and 64 for three, and the number rises very quickly. So DNA stores data with great economy.

### A1.2.10 The code as evidence of common ancestry

pp. 14–24

Almost all organisms use the same genetic code. The chance of this arising separately many times is extremely small, so the best explanation is **universal common ancestry**.

© 2026 Akash Parthi Bendran. Please ask before using or sharing: see whatthecell.org/about.html#permission